



Generative AI for Engineering Managers

The Context Engineering Curriculum

A five-part masterclass that teaches leaders a framework for Generative AI — one that outlives the next model release.

PREPARED FOR	DETAIL
Client	Walmart Global Tech
Audience	Senior managers and leaders — no coding background required
Facilitated by	Dr. Darshan Ingle
Delivery partner	upGrad Enterprise
Format	Five sessions, delivered as half-days

EXECUTIVE SUMMARY

Most executive AI training teaches tools. Six months later the tools have changed and the learning is stale.

This masterclass teaches a framework instead. Leaders learn that a Generative AI system is only as good as the context you give it — and that there are exactly four ways to give a model context, forming a ladder from the cheapest and fastest to the most powerful and most expensive. Learn the ladder, and every AI decision that crosses a leader's desk becomes answerable: which rung does this need, what will it cost, and what could go wrong?

It is built for people who decide, fund, and govern AI — not build it. No coding. Every idea is taught at decision altitude and shown live: a capability running, and a real build opened up so leaders see what it actually takes. Cost sits inside every decision, not in a lecture at the end. This proposal sets out the outcomes, the full day-by-day curriculum, the questions each day equips a leader to answer, and how it is delivered.

Who it is for — and what they leave with

Senior managers, functional heads, and leadership teams who will sponsor and steer AI in their organizations. No prior AI or technical experience is assumed. By the close, each participant has:

- **A working mental model of Generative AI.** clear enough to explain to their own teams and board
- **A decision framework — the Context Ladder.** for choosing the right approach to any use case, and defending the choice
- **Five take-home frameworks.** one per day — practical models they can put to work on Monday morning
- **Economic literacy.** enough to challenge a budget, a vendor quote, or an ROI claim with confidence
- **One of their own use cases, scoped.** mapped to the right rung, costed, and board-ready

The spine — the Context Ladder

One mental model runs through the whole programme. Each day climbs one rung, so nothing feels disconnected — and every buzzword finds its place on a single map.

Rung	Stage	What it lets a leader do
Foundation	How the model works	Read AI clearly — what it is, where it breaks
Stage 1	Prompt Engineering	Get results with instructions alone — cheapest first
Stage 2	RAG + MCP	Bring your own knowledge and systems to the model
Stage 3	AI Agents	Let it plan and act — with guardrails
Stage 4	PEFT / Fine-tuning	Teach the model itself — the last, costliest resort

The one rule. *Climb only when the rung below cannot do the job. Every step up adds cost, complexity, and risk — so you earn each one.*

How it is taught

Every day runs the same four-beat rhythm, so leaders always know where they are on the ladder:

- 1 Frame** — the business problem this rung solves
- 2 Understand** — the concept, in plain language — no code
- 3 See it work** — a live demonstration of the capability

4 Decide — when to use it, what it costs, what the risk is

From Day Three, the live moments come in two forms — both watch-or-try, never a coding exercise:

SEE IT WORK the capability running, in a tool the managers hold themselves — the outcome they would see.

UNDER THE HOOD a real, working build opened up and its logic walked through — so leaders see what a solution actually takes, and can judge it.

The programme at a glance

Five days, each a rung on the ladder and each with its own take-home framework. The full detail follows.

#	Day	Framework	After it, a leader can...
01	Foundations & the Context Ladder	The 4 C's of Model Choice	Talk about AI without bluffing
02	Prompt Engineering	The CRAFT Prompt	Know when a prompt is enough
03	RAG + MCP, and your first assistant	Ground · Connect · Converse	Put your knowledge into AI
04	AI Agents	The Autonomy Dial	Judge where autonomy pays off
05	PEFT + Economics, Delivery & Governance	The AI Investment Canvas	Cost, ROI & KPIs for any decision

THE PROGRAMME, DAY BY DAY

Each day sets out why it belongs in the programme, exactly what is covered, the framework leaders take home, and — most importantly — the questions they will be able to answer by the end of it.

Day One Foundations & the Context Ladder

FOUNDATION · HOW THE MODEL WORKS

Why this day is here. Every later decision rests on understanding what these systems really are and where they fail. Leaders who skip this end up either over-trusting AI or dismissing it — both expensive. Day One builds the shared language, and hands over the map the rest of the programme follows.

WHAT WE COVER

- 1 What Generative AI is** — what a large language model is actually doing when it answers you
- 2 How models are made** — pre-training, fine-tuning, and RLHF, at leadership altitude
- 3 The model landscape** — ChatGPT, Claude and Gemini — and how to compare them for a real decision
- 4 Where models break** — hallucination, bias, and the limits you must design around
- 5 The Context Ladder** — the four ways to shape a model, and the one rule for choosing between them

THE FRAMEWORK The 4 C's of Model Choice

A one-page lens for choosing — or challenging — any model decision, plus the master map: the Context Ladder.

Capability · Cost · Control · Compliance

AFTER DAY 1, A LEADER CAN CONFIDENTLY ANSWER —

- › “What is a large language model actually doing when it answers me?”
- › “Why does it confidently make things up — and can we prevent it?”
- › “ChatGPT, Claude or Gemini — how do we choose, and does it even matter?”
- › “Is our data safe when staff paste things into these tools?”
- › “What do all these buzzwords — RAG, agents, fine-tuning — actually mean?”
- › “Which of the four approaches does my problem actually need?”

SEEN LIVE

SEE IT WORK The same task run across ChatGPT, Claude and Gemini, side by side — where each shines and where each stumbles.

You leave with. a shared vocabulary for AI across your leadership team, and a completed 4 C's scorecard comparing the leading models for your context.

Day Two Prompt Engineering

STAGE 1 · THE CHEAPEST LEVER

Why this day is here. Most “we need an AI project” problems are actually solved by a better instruction — at almost no cost. Leaders who can tell a strong prompt from a fragile one stop their organisations buying heavy solutions to light problems.

WHAT WE COVER

- 1 The anatomy of a prompt** — zero-shot, few-shot, and chain-of-thought, in plain terms
- 2 Structured outputs** — answers you can trust and drop straight into a workflow
- 3 Multimodal prompting** — working across text, images, and documents
- 4 Secure prompting** — prompt injection and the failure modes a leader must know

THE FRAMEWORK The CRAFT Prompt

A repeatable recipe anyone on the team can use to get reliable results from any model.

Context · Role · Ask · Format · Test

AFTER DAY 2, A LEADER CAN CONFIDENTLY ANSWER —

- › “When is a well-written prompt all we need — and when do we need more?”
- › “Why do we get a different answer every time, and how do we make it consistent?”
- › “Can we trust it to read our contracts, invoices, and images?”
- › “What is prompt injection, and is it a risk we should worry about?”
- › “How do we turn one good prompt into something the whole team reuses?”
- › “Can this replace a manual, rules-based process we run today?”

SEEN LIVE

SEE IT WORK Leaders write, break, and refine a prompt for a real task from their own world — live, in ChatGPT.

UNDER THE HOOD how a single reusable prompt template gets wired into a repeatable workflow.

You leave with. the CRAFT template, and the start of a prompt library for your own function.

Day Three RAG + MCP, and your first assistant

STAGE 2 · YOUR KNOWLEDGE, YOUR SYSTEMS

Why this day is here. A model out of the box knows nothing about your business and can't touch your systems. This is the day AI stops being a clever toy and starts working with your knowledge — the single biggest unlock for most organisations, and the one most often done wrong.

WHAT WE COVER

- 1 Data strategy first** — why a model doesn't know your business, and the data foundation retrieval depends on
- 2 Retrieval-Augmented Generation (RAG)** — grounding answers in your own documents, explained without the math
- 3 Tool calling & APIs** — how a model reaches your systems — APIs explained here, as the pieces that connect
- 4 The Model Context Protocol (MCP)** — the emerging standard for wiring AI to your tools safely, and why it matters now
- 5 Chatbots, after RAG** — conversation design and memory — turning retrieval into an assistant people trust

THE FRAMEWORK Ground · Connect · Converse

The three moves that turn a raw model into your assistant — plus the Knowledge Test: fresh, private, needs a source? Retrieve, don't retrain.

Ground it (RAG) · Connect it (Tools · MCP) · Converse (Chatbot · Memory)

AFTER DAY 3, A LEADER CAN CONFIDENTLY ANSWER —

- › “Can AI answer using our own policies, contracts, and product docs?”
- › “How do we stop it inventing answers — can it show its sources?”
- › “What is MCP, and why does it change what's possible for us?”
- › “Should we build a company assistant — and what does it actually take?”
- › “Do we need to fine-tune the model, or is RAG enough? (usually, it's enough)”
- › “What data do we need in order before any of this can work?”

SEEN LIVE

SEE IT WORK A RAG assistant answering real questions from a company's own document set — with its sources shown.

UNDER THE HOOD a real HR-assistant chatbot opened up — retrieval, memory, and a chat interface — and the logic behind each choice.

You leave with. the Ground-Connect-Converse blueprint, and a shortlist of your own best first-assistant candidates.

Day Four AI Agents

STAGE 3 · AUTONOMY & ORCHESTRATION

Why this day is here. Agents are where the hype is loudest and the risk is highest. Leaders need to separate what agents genuinely do from what vendors claim — and know exactly which controls keep autonomy safe before they fund it.

WHAT WE COVER

- 1 What an agent actually is** — reasoning loops, planning, and taking action
- 2 Orchestration** — how multi-step work gets coordinated — the concept, not the plumbing
- 3 Keeping agents safe** — reliability, guardrails, and human-in-the-loop by design
- 4 Agent risk** — where autonomy goes wrong, and the security questions to ask

THE FRAMEWORK The Autonomy Dial

Autonomy is a dial, not a switch. Leaders learn to set it deliberately — each notch trades more speed for more risk — with a guardrail checklist for each.

Assist · Draft · Act w/ approval · Act & report · Autonomous

AFTER DAY 4, A LEADER CAN CONFIDENTLY ANSWER —

- › “What is the real difference between a chatbot and an agent?”
- › “Where does autonomy save us real money — and where does it just burn tokens?”
- › “What could go wrong if we let AI act on its own, and how do we contain it?”
- › “What controls must be in place before an agent touches a live system?”
- › “Can it answer questions straight from our database in plain English?”
- › “How much human oversight is enough — and where must a human stay in the loop?”

SEEN LIVE

SEE IT WORK An agent completing a multi-step task end to end — and exactly where it hands back to a human.

UNDER THE HOOD a natural-language-to-SQL agent (ask in plain English, watch it write and run the query) and a working agent from an enterprise build.

You leave with. the Autonomy Dial and guardrail checklist, applied to one candidate process of your own.

Day Five PEFT + Economics, Delivery & Governance

STAGE 4 · THE LAST RUNG, AND THE WHOLE PICTURE

Why this day is here. This is where understanding becomes decisions a leader can defend — the cost, the ROI, the KPIs, and the governance a board will ask about. It is also where the pilot-to-production gap is crossed, or not.

WHAT WE COVER

- 1 Fine-tuning and PEFT** — teaching the model itself — what it takes, and why it's the last resort
- 2 From pilot to production** — why most pilots stall, and how AI apps get built and shipped — web apps and CI/CD shown as options, with the logic
- 3 The Economics of AI** — token economics, model routing, batching, and caching — the true cost of each rung
- 4 ROI, and build vs. buy vs. partner** — AI vs. rule-based vs. human, the vendor-strategy call, and the business case
- 5 Governance, guardrails & KPIs** — evaluations, monitoring, guardrails — and the KPIs that tell you it's actually working

THE FRAMEWORK The AI Investment Canvas

One page a leader can take to any budget conversation — plus the Pilot-to-Production Gap: the five things that quietly kill pilots.

Value · Cost of Ownership · Risk & Governance · KPIs · Build / Buy / Partner

AFTER DAY 5, A LEADER CAN CONFIDENTLY ANSWER —

- › “When is fine-tuning worth it — and when is it money down the drain?”
- › “Why do most AI pilots never reach production, and how do we beat the odds?”
- › “What will this actually cost us per month, at scale?”
- › “Do we build it, buy it, or partner — and how do we choose a vendor?”
- › “How do we know it's working? What belongs on the dashboard?”
- › “How do I defend this investment to the board?”

SEEN LIVE

SEE IT WORK Capstone — each leader maps one of their own use cases to the right rung, with a cost, an ROI, and the KPIs to defend it.

UNDER THE HOOD a finished AI product and the pipeline that ships it — presented as options, with the logic behind each.

You leave with. your own use case on the AI Investment Canvas — a board-ready one-pager.

THE TECHNOLOGY — A BROWSER IS ALL THEY NEED

The entire programme runs in a web browser. Nothing is installed on any laptop — which keeps corporate IT out of the critical path. Participants touch one familiar tool for the hands-on work; everything more advanced is shown in pre-built notebooks the facilitator runs.

What happens	The tool participants use	Setup
Chat, prompt, compare answers	ChatGPT (Plus) — with Claude & Gemini shown alongside for the model-comparison moment	Browser · a ChatGPT account
Feel RAG & a simple agent — no code	ChatGPT Custom GPTs — upload documents for grounded answers; add an Action for a light agent	Browser · same account
See how it is really built	Jupyter notebooks on Google Colab — RAG, tool calling & MCP, an AI agent, a text-to-SQL example, and a light fine-tuning demo	Browser · a Google account · notebooks provided
See a finished product	Live reference builds — a real assistant chatbot, an agent, and a shipped app	Nothing · facilitator-run

WHY THIS STACK

Nothing to install (corporate laptops stay untouched), a front door they already know in ChatGPT, and watch-or-try optionality on every demo. Managers experience Generative AI without ever being asked to write a line of code.

What each participant needs

- **A laptop and a modern browser.** Chrome, Edge, or Safari — nothing else installed
- **A ChatGPT account.** the free tier works; ChatGPT Plus is recommended for the full hands-on experience
- **A Google account.** to open the Colab notebooks the facilitator shares
- **No coding, no prior AI experience.** just a decision-maker's curiosity

Delivery & logistics

ITEM	DETAIL
Format	Five sessions, delivered as half-days — five parts in total
Mode	In-person, live-virtual, or hybrid — your choice
Cohort size	Up to [20–25] participants for full interaction
Cadence	Weekly, or an intensive block — agreed with you
Materials	All Colab notebooks, a reusable prompt library, the five frameworks, and a leave-behind after each day
Language	English

Data, privacy & running costs

All demonstrations use synthetic or sample data — never your confidential information — unless you specifically request a private pilot on your own material. For organisations that want participants working with internal data, we recommend running the hands-on work inside your own ChatGPT Enterprise or Team workspace.

A TEACHING MOMENT

The model-API usage behind the live demos runs on capped keys the facilitator provides. The real token cost is shown on screen during the economics session — so the running cost of the course becomes part of the lesson on the cost of AI.